Network administration

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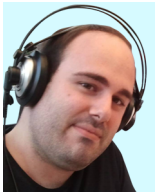

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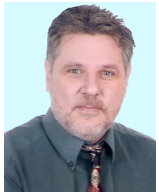

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Introduction

When designing and deploying a network, keep these six basic steps in mind:

1) Select the cabling and hardware design.
2) Install the hardware and network operating system.
3) Configure the operating system and load the applications.
4) Create the user environment.
5) Start up network administration.
6) Maintain and monitor network activity.

See also

See the *User administration* module.

This module provides a brief overview of various aspects of computer network design and deployment. It begins by describing the components that make up a network and ends with a broad outline of the steps involved in connecting a workstation to the network. This document is not designed to serve as a comprehensive reference on this extensive and multifaceted topic. Instead, it aims to offer administrators foundational guidelines to assist them in initiating and maintaining progress on tasks of this complexity.

Objectives

The learning resources in this module cover the content and key tools needed to achieve the following objectives:

1. To be able to describe the basic hardware components of a computer network and understand how these parts are connected to each other. Based on this description, the appropriate devices and topologies for the services that the network must provide can be identified.

2. To learn about the communication protocols that must be used by the computers on the network and also how the workstations and some services are configured. In general, the actions that must be carried out are similar, regardless of the network architecture and operating system.

3. To describe the tasks that the administrator must carry out once the network is up and running (maintenance, monitoring and security tasks).

1. Getting started with network administration

Personal computers can give people control over their own data to do what they need. However, standalone computers can't give you direct access to different data across an organisation, or easily share their information or programs. Networks provide a middle ground between these two extremes: individual and centralised processing.

There are many benefits to deploying a network, including the following:

- **Sharing peripheral devices**: users across the organisation must be able to access shared resources such as mass storage devices, high-cost output devices (e.g. laser printers or plotters) and any other devices.

- **Organisation-wide communication**: users in the organisation must be able to communicate with each other, whether by email, video calls, corporate instant messaging or any other tool, to be able to do their work properly.

- **Easy software maintenance**: a large part of the software is often shared over the network, rather than being installed individually on each workstation.

- **Centralised shared resource management**: for the organisation to function properly, the ability to share resources (tools, data, servers, etc.) is very important, regardless of how geographically dispersed the organisation may be. Organisations often do business in different cities or even countries. Nevertheless, resources must be accessible at all locations.

Depending on this geographical spread, networks can be classified as follows:

- **LAN (local area network)**: a network that connects computers in a relatively small, predefined area measuring between 10 and 1,000 m (e.g. an office, a campus, etc.).

- **MAN (metropolitan area network)**: a network that uses a high-speed connection to provide coverage in a large geographical area of 1 to 10 km (e.g. a city).

- **WAN (wide area network)**: networks that connect cities, countries and continents and provide links between locations more than 10 km apart (e.g. a country).

Considering how important a network is to any organisation's day-to-day operations, it is mandatory to have someone who can design, deploy, maintain, and update it. This role is fulfilled by the network administrator, who must be familiar with the network's hardware, the communication protocols between the different computers (and their operating systems), and the minimum security requirements to be met by the network.

2. Physical components and network design

Before designing a network, there are a few basic aspects to consider, as outlined in the following questions:

- How many computers (workstations) need to be connected to the network?
- How many computers will need to be added in the future?
- Where and how will the computers be located? (This requires a diagram showing the distribution of the machines).
- Do we need a server?
- What connection speed is required?
- What resources need to be shared?
- What software do we want to install? Do we have the version that can run on a network?

Before answering these questions, the following concepts from other computer networking courses should be reviewed.

2.1. Network components

A network's basic components include cabling, connection points, and interconnection points.

2.1.1. Network cabling

There are several types of cables:

1) Twisted pair: this type of cable consists of several conductive wires twisted together to protect them from surrounding noise. They are the most economical and easiest type of cable to install. They have a range of up to 100 m (without signal loss) and speeds between 10 and 100 Mbps.

There are several categories of twisted pair cable:

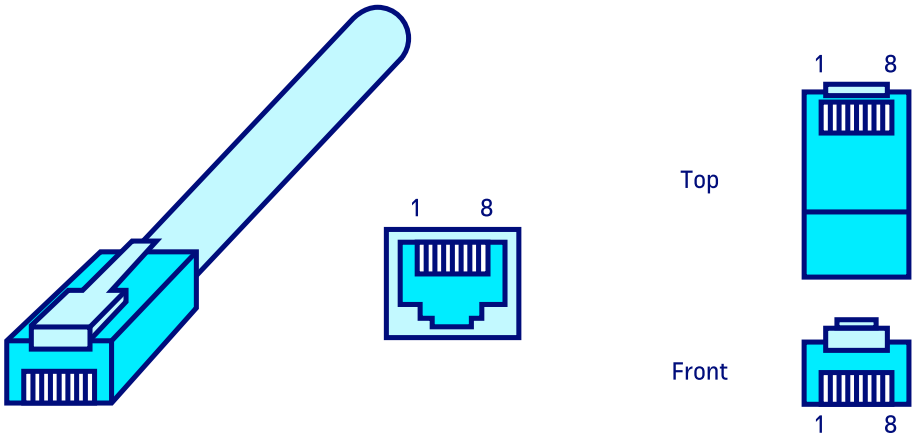
- **STP (shielded twisted pair)**: shielded cable, consisting of two pairs of conductive wires covered by a mesh.

- **UTP (unshielded twisted pair)**: unshielded cable, consisting of four pairs of conductive wires. Figure 1 shows what a UTP twisted pair looks like (the female and male connectors, respectively). UTP cables can also be broken down into different categories based on their transmission speed:
  - **Category 3**: can reach transmission speeds of 30 Mbps.
  - **Category 5**: the most commonly used type of cable. It can reach speeds of 100 Mbps (Fast Ethernet networks). Category 5a (also called *5+* or *5e*) improves on Category 5 and can reach speeds of up to 1,000 Mbps (Gigabit Ethernet networks).
  - **Category 6**: can reach speeds of 1,000 Mbps.

UTP categories

Categories 6e and 7 also exist and are intended for future use in 10 Gigabit Ethernet (10,000 Mbps) networks.

Figure 1. What a UTP cable looks like



By connecting all computers to a hub using twisted pair cables instead of a dedicated server, it is possible to design a very simple network that is perfectly suited for sharing resources and can be easily expanded to occupy all the hub ports.

Peer-to-peer

Networks without a dedicated server are called *peer-to-peer networks*. All workstations have the same status and share resources.

Crossover

When using a hub, both ends of the cable (one connected to the hub and the other connected to the computer's network card) are inserted into the RJ45 connector in the same way. However, when the two ends are connected directly between two computers, a crossover cable is required to swap the order.

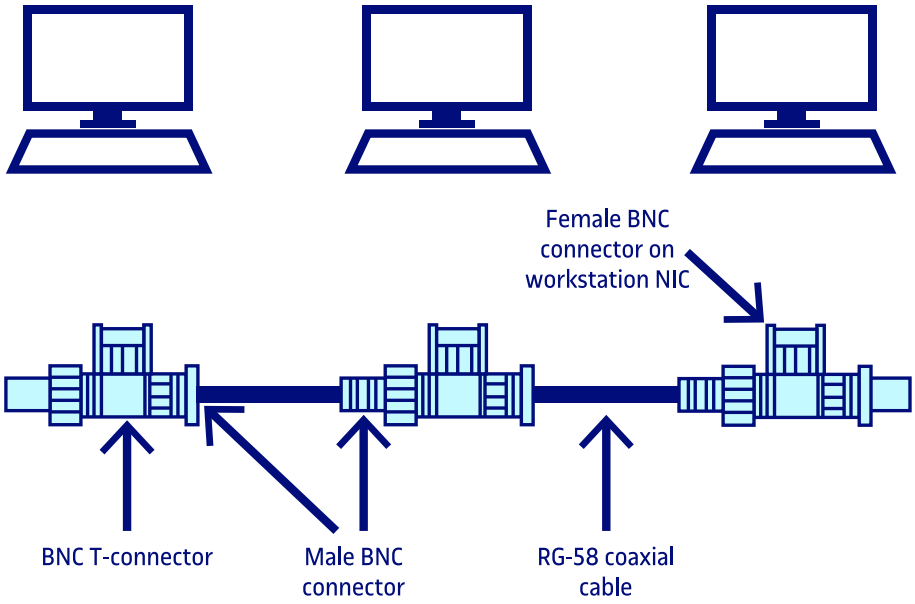
2) Coaxial cable: this cable has a single internal conductor and multiple layers of protection. It is available in thick and thin versions (RG-58A/U) for distances of up to 2 km, where it can achieve transmission speeds of 20 Mbps, while over short distances (no more than 100 m) it can reach 100 Mbps. Unlike twisted pair cables, coaxial cables reduce signal damping over long distances and minimise the percentage of power lost through radiation. They are highly susceptible to espionage and noise from electrical devices (e.g. engines).

See also

See the *Security administration* module.

BNC connectors are used to connect different sections of coaxial cable, while T-shaped BNC connectors are used to connect a computer to the network, as shown in Figure 2.

Figure 2. How to connect a computer to the network with a coaxial cable



3) Fibre optic: information is transmitted by a beam of light running through a photoconductive core. It provides high bandwidth and can reach transmission speeds of hundreds of Mbps or even Gbps. Fibre optic cables suffer minimal signal damping, are immune to electromagnetic interference and difficult to intercept and spy on, since they do not emit any monitorable signals. They are normally used together with other types of cabling. There are two different types of fibre optic cable: single-mode and multi-mode. The former only supports a single mode of transport (they can only transmit light beams that follow the axis of the fibre). They have a bandwidth that can reach 100 GHz/km. Multi-mode fibre optic, with a larger core diameter than single-mode fibre optic, can carry multiple modes simultaneously. It is easier to install and has a bandwidth of up to 500 MHz/km (lower than single-mode fibre optic). This makes it particularly suitable for video surveillance systems or LANs, for example.

2.1.2. Network connection components

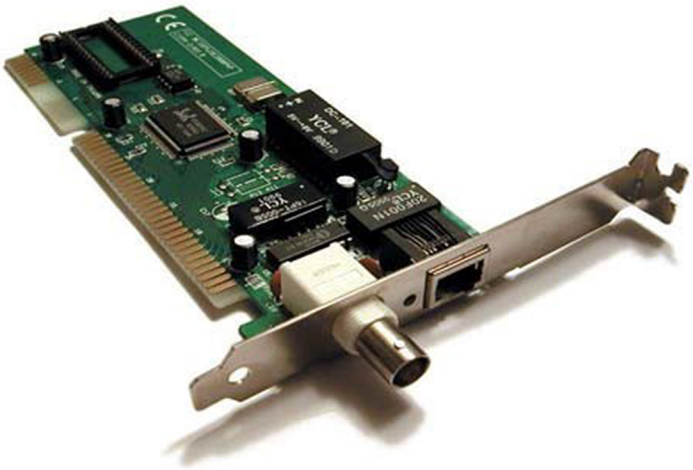
The following network connection components exist:

1) Network interface cards (NIC): computers are connected to the network via network interface cards. Figure 3 shows what a NIC looks like.

Network interface cards

These cards are usually found in PCI, or PCMCIA format for laptops. An increasing number of laptops and desktop computers have a built-in card on the motherboard. Some hand-held devices have a NIC for connecting to a wireless local area network.

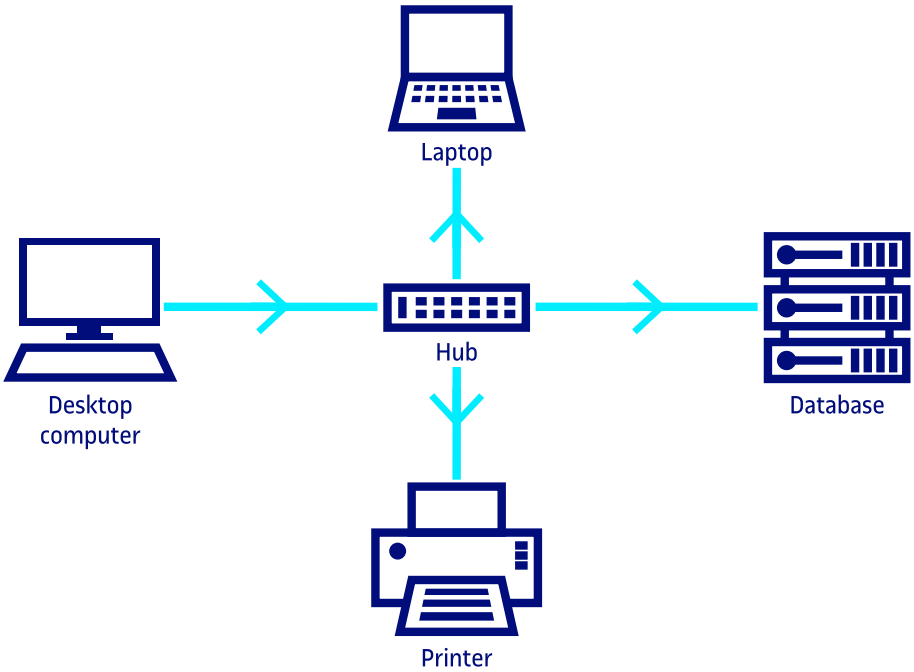
Figure 3. Network interface card



2) Firewall: any device (hardware or software) that prevents unauthorised users from accessing a specific machine.

3) Hub: as shown in Figure 4, these are devices that share a communication line between several computers. They distribute all the information they receive to all devices connected to their ports.

Figure 4. Hub configuration example



Bandwidth

All workstations connected to the same hub or hub stack compete for channel bandwidth.

4) Switch: manages network traffic flow based on each packet's destination address. Therefore, switches can determine which devices are connected to their ports and redirect information only to the destination port, unlike hubs, which send information indiscriminately.

5) Backbone: the main cables that connect the segments of a local network are called the backbone. They are usually high-speed links (e.g. fibre optics).

6) Switch cabinets: the network is generally divided into different switch cabinets that cover the entire network service in a given environment, such as an entire floor of a building. All these cabinets are connected to a central cabinet where all communications are usually grouped together and the different servers are installed. This is usually a room that is properly climate-controlled, in terms of both temperature and humidity, and normally has an uninterruptible power supply.

7) Server: a computer that shares its peripherals with other computers on the network. There are many different types and can be grouped into three general categories: print servers, communications servers and file servers. Within a network, they may be used exclusively to provide these services, or they may also be used as workstations.

8) Workstation: each workstation runs its own operating system (Unix, Linux, Windows, etc.) and this operating system runs network software that is used to communicate with the servers and other devices on the network, simplifying the management of both local and server resources.

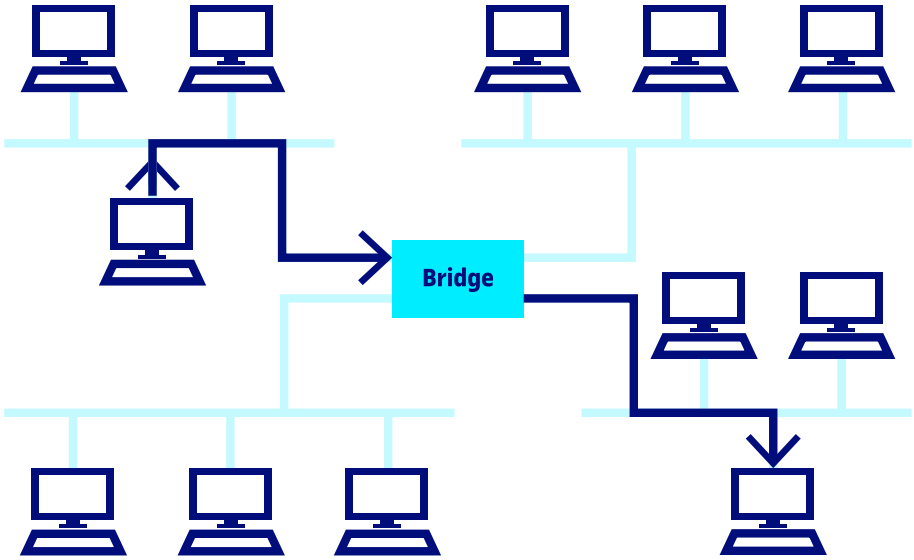
2.1.3. Network interconnection components

There are four different types of interconnection components:

1) Repeaters: these are 'non-smart' devices that amplify signals and prevent damping problems that occur when cables reach a certain distance (remember that these distances vary depending on the type of cabling used). They operate on what is known as the physical layer.

2) Bridge: these connect two network segments (which may be different, see the example in Figure 5) and operate at the data link layer. Unlike repeaters, bridges are 'smart' enough to filter information traffic between segments and can self-learn, filter and forward the information that passes through them. When a bridge is added, each segment has a different IP address, so information is always routed to its destination and bottlenecks are avoided when all workstations are connected to the same segment.

Figure 5. Network interconnection via a bridge



3) Router: devices that manage packet traffic flowing from outside the network to the inside (and vice versa). They can be highly sophisticated devices and can act as firewalls. Similar to bridges, they offer data routing services, i.e. they can filter information and find the most efficient destination route for the packets being transmitted.

4) Gateway: these operate at the upper levels of the OSI protocol hierarchy. They connect networks that use incompatible protocols.

2.2. Network topology and types

Network topology refers to the physical pathway taken by data through the network, i.e. the logical way different devices are connected. A distinction is often made between **logical topology** and **physical topology** or **layout** (how the wiring is performed).

There are basically three topologies to consider in a LAN:

1) Bus topology: in a bus network, all nodes (servers and workstations) are connected to a common cable (bus), as shown in Figure 6. The most common features of this topology are listed below:

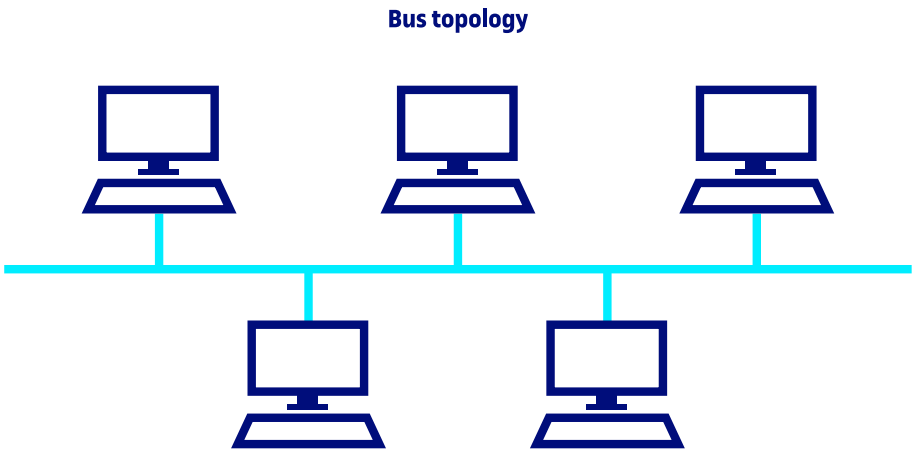
Logical topology

Logical topology may differ from topography, as we will see later.

- Nodes do not retransmit or amplify information.
- Information retention time in nodes is zero.
- All messages reach all nodes.
- No information routing is required.
- Communication reliability depends solely on the bus (critical point).
- Configuring is flexible and modular.
- Low-cost technology that is still frequently used.
- Information circulating can be easily intercepted.

With only a single bus, overload traffic can lead to a significant drop in network performance. To control network traffic, switches capable of discriminating between different types of traffic can be used.

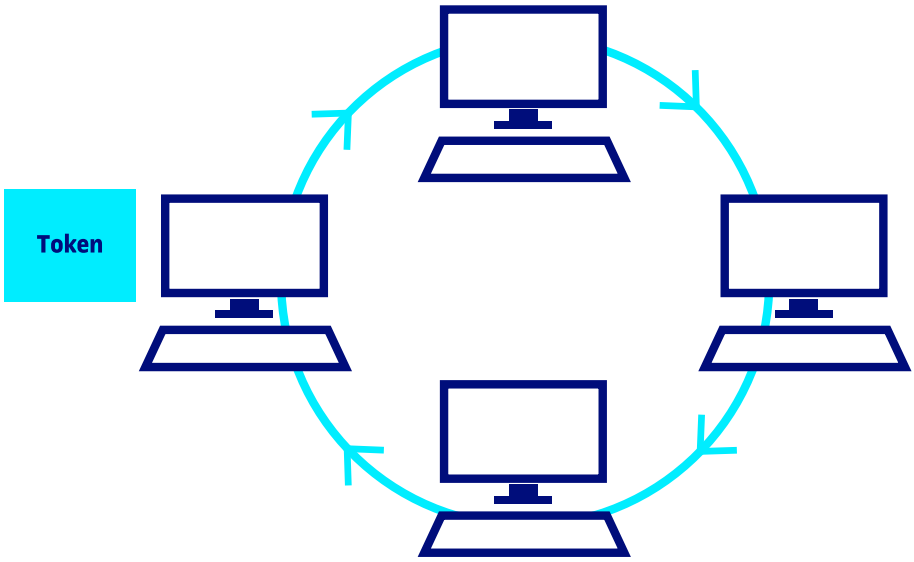
Figure 6. Bus topology network



2) Ring topology: the cable runs from workstation to workstation (and to the server) without any end points in a ring network. Each node has connections to two other workstations. This type of network can be seen in Figure 7. The most common features of this topology are listed below:

- Each node amplifies and distributes the information it receives.
- Messages are sent from node to node along the ring, which means that all information passes through all the communication modules at the workstation (making it easy to intercept information).
- Information routing is not necessary.
- The ring's reliability depends on each of the nodes and the communication path forming the ring. The failure of a single workstation could cause the entire network to crash.

Figure 7. Ring topology network



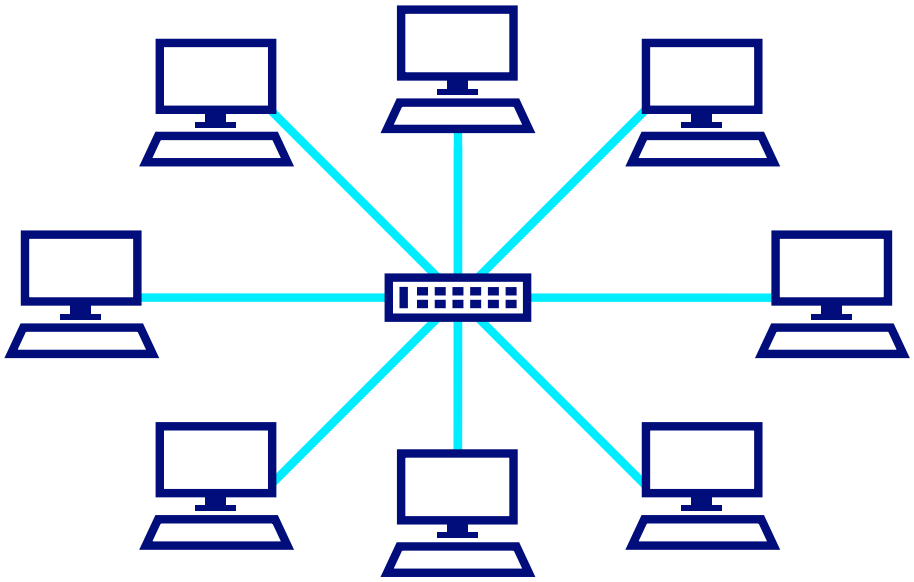
3) Star topology: as shown in Figure 8, all workstations and the server are connected to a single hub or switch. The most important difference between this topology and the others is how the connections are centralised. This feature gives it extra resilience to workstation crashes, but its main drawback is that it has a critical point, the central component, which, if attacked or crashes for any reason, can cause the entire network to go down. The most common features of this topology are listed below:

- All workstation communicate with each other via a central node.
- The central device can be active or passive.
- Faults have varying degrees of impact depending on where they occur.

Network maintenance

"Documenting" the design and components that make up the network helps with future maintenance tasks.

Figure 8. Star topology network



When choosing a network topology, the following considerations should be made:

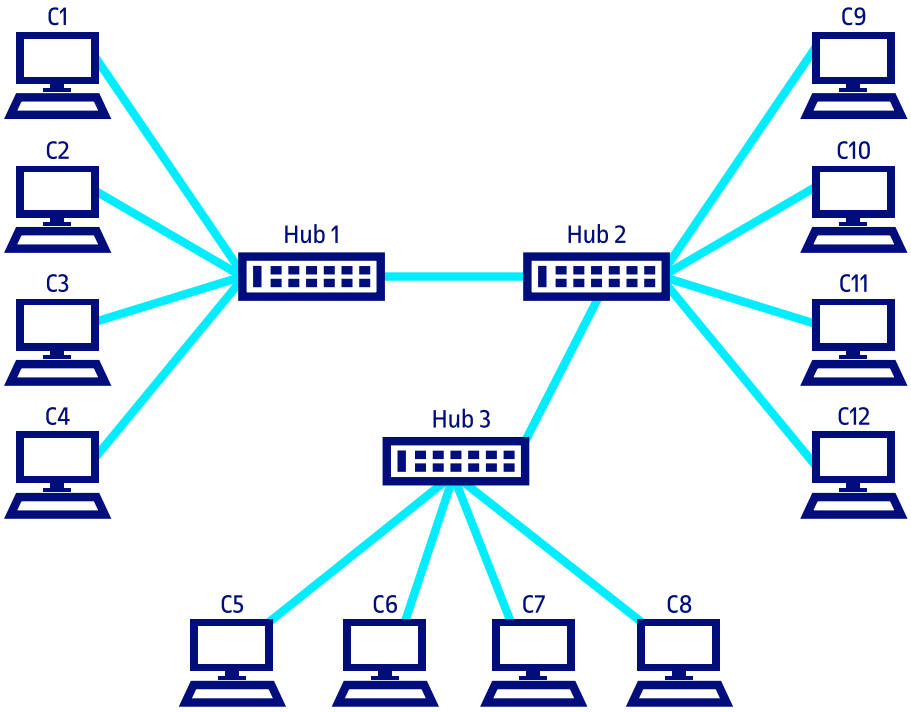
- Maximum achievable distance.
- Maximum number of workstation.
- Flexibility to add or remove workstations.
- Workstation fault tolerance.
- Message delay.
- Cost.
- Information flow through the network.

Cable positioning

Although they are not physically connected, the cables should be 10% longer than the expected length.

Bus and ring topologies were the most commonly used in local networks. However, due to flexibility, reliability, and security reasons, the star topology has become the most popular. In networks where a bus or ring topology would be the most logical choice for the workstation layout, a physical star topology is often used instead.

Figure 9. Hub-based physical layout



As shown in Figure 9, all signals from all workstation are combined in the hub and transmitted to all workstation as if it were a bus configuration. Although this generates high traffic, costs and maintenance issues are reduced.

2.3. Local network types

The Institute of Electrical and Electronic Engineers (IEEE) was founded in 1980 and developed the **IEEE 802.x standards**, defining standard operating procedures for local area networks.

The IEEE 802.x standards define local area network operating standards. The most commonly used standards are described below:

- **IEEE 802.3**: standard based on Ethernet network version 2.0. It defines a network with bus topology and CSMA/CD access method (all workstation can access the medium simultaneously and compete for use of the communication channel). It is used in technical environments and offices, universities and hospitals.

- **IEEE 802.4 (obsolete)**: defines a network with bus topology and token passing. Only the workstation with the token can access the channel. It was used in industrial environments and is known as Token Bus.

- **IEEE 802.5 (obsolete)**: standard based on IBM's Token Ring network. It defines a network with a ring topology and token passing. It was used in office environments.

- **IEEE 802.15**: defines wireless personal area networks, used for interconnecting personal devices. It is also known as Bluetooth.

- **IEEE 802.16**: defines broadband wireless access networks known as WiMAX, which are designed to provide wireless home access to the network.

The **Ethernet networks** are probably the most popular of all the standards mentioned above.

How to make a major technological change

If a major technological change is required (for example, switching from 10 Mbps Ethernet to 100 Mbps or 1 Gbps), consider how best to proceed: gradually or all at once, interrupting all services for as long as necessary.

Most Ethernet network deployments have transmission speeds of 10 Mbps. Below are the **different types according to the cabling** used (the first number refers to the speed in Mbps and the second to the metres that the segment can have – multiplied by one hundred – without the signal losing strength):

- **1Base-5**: twisted pair cable with a transmission speed of 1 Mbps and a maximum segment length of five hundred metres.

- **10Base-T**: UTP twisted pair cable with a maximum segment length of 100 metres in a star physical topology.

- **100Base-T**: similar to the above, but with transmission speeds of 100 Mbps (also called *Fast Ethernet*).

- **10Base-5 (thick wire)**: thick coaxial cable with a transmission speed of 10 Mbps. It can handle up to one hundred workstations in segments ranging from a few metres to five hundred metres long.

- **10Base-2 (thin wire)**: thin coaxial cable with a transmission speed of 10 Mbps. It can handle up to thirty workstations in segments of up to one hundred and eighty-five metres long.

- **10Base-F**: fibre optic with transmission speeds of 10 Mbps.

2.3.1. Wireless local area network (WLAN)

When choosing a network type, it is worth considering other options besides the types we have looked at so far. In all the types examined, we have identified the following problems:

- The difficulty, or even impossibility, of running cables when the site is physically difficult to access.

- The need to estimate network growth and develop more infrastructure than is initially required to accommodate this future growth.

- All options require cutting holes in walls or floors to feed the necessary cables.

WLANs (wireless local area networks), i.e. wireless local networks based on radio or infrared waves, have emerged as a solution to this type of problem. The primary objective of these networks is end-user convenience (i.e., the ability to connect to the network from anywhere in the organisation at any time), followed by ease of deployment and network growth (without forgetting that aspects such as reliability and bandwidth are also important).

The IEEE has defined the 802.11 standard (and subsequent standards) to regulate how wireless networks work. The most common is the 802.11ax standard, with speeds of up to 10 Gbps. It transmits using the 2.4 and 5 GHz frequency bands (*industrial, scientific and medical*).

The main features of these standards are shown in Table 1.

Table 1. Wi-Fi standards

Protocol	Standardisation date	Frequencies	Speed (typical)	Speed (maximum)	Indoor range	Outdoor range
Initial standard	1997	2.4-2.5 GHz	1 Mbps	2 Mbps	?	?
802.11a	1999	5.15-5.35 GHz 5.47-5.725 GHz 5.725-5.875 GHz	25 Mbps	54 Mbps	~25 m	~75 m
802.11b	1999	2.4-2.5 GHz	6.5 Mbps	11 Mbps	~35 m	~100 m
802.11g	2003	2.4-2.5 GHz	25 Mbps	54 Mbps	~25 m	~75 m

Protocol	Standardis-ation date	Frequencies	Speed (typical)	Speed (max-imum)	Indoor range	Outdoor range
802.11n	2009	2.4 GHz and/or 5 GHz	200 Mbps	450 Mbps	~50 m	~125 m
802.11ad	2012	2 Ghz, 4 Ghz, 5 Ghz, 60 Ghz	1,500 Mbps	7.0 Gbps	~10 m	~30 m
802.11ac	January 2014	5.15-5.35 GHz 5.47-5.875 GHz	433 Mbps	1,300 Mbps	~20 m	~50 m
801.22af	February 2014	TV bands between 54-790 MHz	50 Mbps	568.9 Mbps		~1,000 m
802.11ax	2019	2.4 GHz and/or 5 GHz	600 Mbps	9.6 Gbps	~30 m	~120 m
802.11be	2024	2.4 GHz, 5 GHz, and/or 6 GHz	2 Gbps	46 Gbps	~30 m	~120 m

2.3.2. Wi-Fi protected access (WPA)

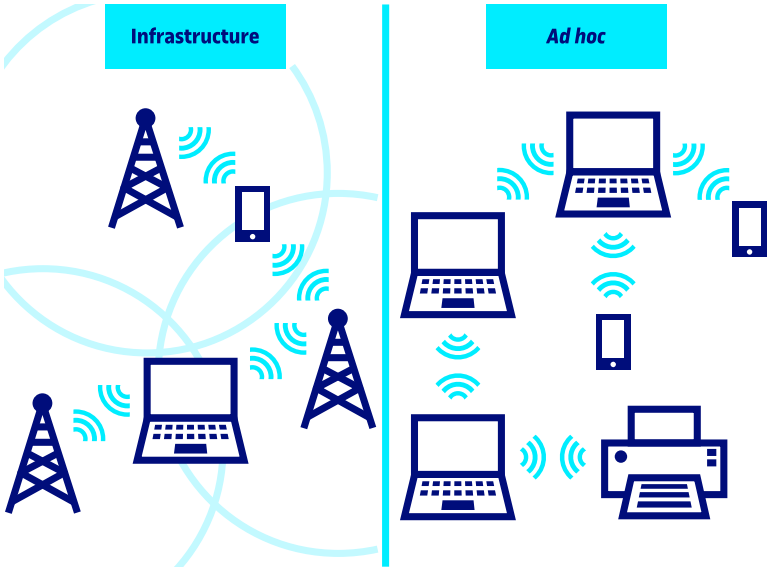
Given that information does not need any specific means to circulate, these networks pose significant security issues. For example, the firewall is usually a critical security element in a normal network configuration and provides most of the protection against external attacks. In a wireless network, attackers no longer need to 'break through' the firewall and can attack other devices on the network directly. The 802.11 standard initially defined the WEP (wired equivalent protocol) to solve these problems. However, it is not a secure protection mechanism because it can now be easily cracked.

In response to the security issues caused by the WEP protocol, WPA (Wi-Fi Protected Access) was developed, which was initially defined in the 802.11i specification. Therefore, today with WPA3, **security features** such as AES encryption, a better authentication protocol (use of WPA) and message integrity control (use of the MIC hash function instead of the CRC-32 used in the WEP protocol) are available. However, it is important to remember that, due to their intrinsic nature, WLAN networks require greater security measures than those that would be adopted in a normal "wired" network.

2.3.3. *Ad hoc* networks

WLANs can operate in *ad hoc* mode or infrastructure mode:

Figure 10. Infrastructure and *ad hoc* Wi-Fi networks



- **Ad hoc mode (client-to-client)**: all machines within the same range can communicate with each other directly. Although uncommon, this mode is useful for exchanging information between two computers (similar to connecting two computers with a twisted pair cable). An example is shown on the right-hand side of Figure 10.

- **Infrastructure mode (client to access point)**: workstation communicate with access points, which operate as repeaters and broadcast information to the rest of the network, as shown on the left-hand side of Figure 10.

WEP

Wired Equivalent Protocol is based on RC4 encryption. A default WEP key must be set on each access point and on each client. Only clients with the same key are allowed access.

Mobile components

To fully benefit from wireless networks, workstations should also be mobile devices such as laptops or handheld devices.

Finally, WiMAX (worldwide interoperability for microwave access) technology must also be mentioned. This is a wireless data transmission standard (IEEE 802.16) designed for use in metropolitan areas, providing concurrent access in areas with a maximum radius of 48 kilometres and transmission speeds of up to 70 Mbps. This technology clearly makes it possible to connect our mobile devices (laptops, handheld devices, etc.) anywhere and, among other advantages, could provide Internet to hard-to-reach areas where infrastructure cannot be installed. It transmits in the 2 to 11 GHz and 10 to 60 GHz bands for communication between service provider antennas. An AES encryption algorithm is used to protect the transmitted data.

2.3.4. Fibre optic networks

Fibre optic networks use a highly transparent thread made of glass or plastic to send pulses of light that represent the data to be transmitted. The light beam is kept within the fibre and travels by internal reflection. The light source can come from a laser or an LED diode. This cable is widely used and can transmit large amounts of information in a short time and over long distances in communications. It supports extremely high bandwidths and speeds. The

vast amount of information that can be transmitted per unit of fibre optic cable is undoubtedly its greatest advantage. As for cost, fibre optic cable can be produced at a lower cost than copper cable.

Fibre optic networks are widespread in business and home environments. They were first used to connect servers to server racks and to interconnect servers with high data flows, primarily in supercomputing environments and later for interconnecting high-performance, high-availability devices in which different parts of the service (web server, application server, database server) are located. Another typical use of fibre optics has been to create dedicated interconnection lines between different company work centres.

Furthermore, it is increasingly common to use cloud computing, not only for data storage but also for processing (AWS, Azure, etc.). These new computing models require the use of fibre optic networks to transport large volumes of data at extremely high speeds.

In the non-business world, high demand for videocalling and on-demand multimedia content distribution services is driving the popularity of streaming services provided by both communications operators and companies specialising in this type of service (Netflix, HBO, Disney, etc.). Given the large volume of data that must be sent over the network to view content in real time and the high transmission speed required for good service quality (QoS), fibre optic networks are ideal for users of this type of service.

3. Communication protocols

After installing the hardware, cabling and various devices that form the network, you need to install the network software, which will manage all services. These services are organised using a protocol suite that allow different computers on the network to communicate with each other. The most common protocols belong to the TCP/IP family (among others, such as Apple Talk for Apple Macintosh systems, which has been disbanded since the release of Mac OS X v10.6 in 2009, IPX/SPX, etc.).

3.1. TCP/IP

TCP/IP is a protocol suite used for sharing resources between computers on a network. It was developed in 1972 by the United States Department of Defense to link resources on the ARPANET (a Department of Defense network) and, since then, has become widely used on the Internet. It is also closely linked to the Unix operating system, although most operating systems now support TCP/IP. TCP/IP protocols are extremely flexible, which means that almost all underlying technologies (Ethernet, Token Ring, etc.) can be used to transmit TCP/IP traffic.

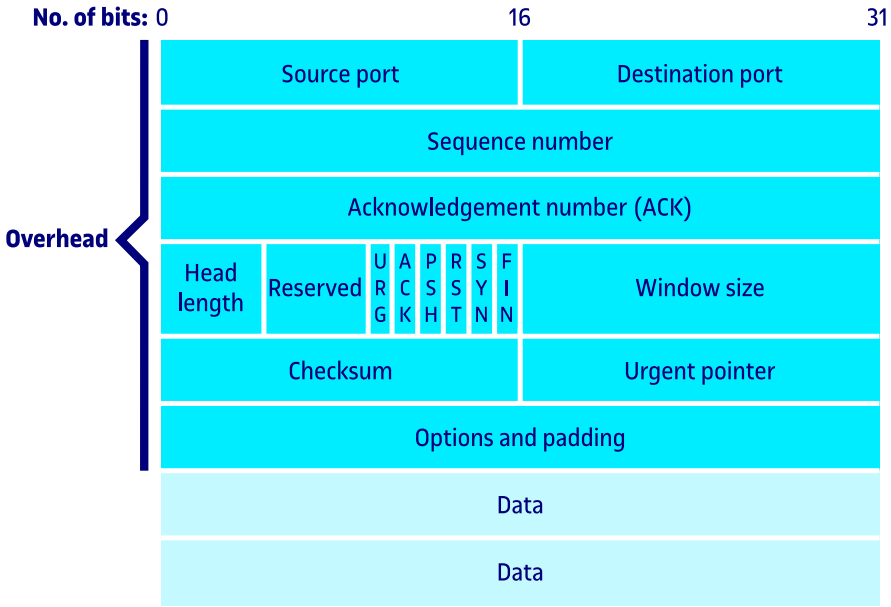
Protocols

TCP and **IP** are only two of the protocols included in the general TCP/IP suite, but they are the most well-known and give the entire set its name.

The **UDP** (user datagram protocol) is an unreliable, connectionless protocol in the transport layer of the OSI model (the same as the TCP protocol).

Other TCP/IP **protocols**: ARP (address resolution protocol), ICMP (Internet control message protocol).

Figure 11. TCP segment



When using the TCP/IP protocol, information is sent as a sequence of **datagrams**, which, as shown in Figure 11, contain the data to be sent and control information. Each of these datagrams is sent individually to the network in such a way that the original information can be reconstructed on arrival at the destination machine by regrouping the datagrams sent (i.e. the datagrams do not necessarily have to arrive in the same order in which they were delivered).

The TCP (transmission control protocol) guarantees that the data is received and that the datagrams are reassembled in the correct sequence (reliable end-to-end transmission service). At the same time, this service operates on top of the IP (Internet Protocol), which is unreliable and performs datagram routing functions.

The TCP/IP architecture has four levels or layers, which group protocols and relate to the OSI levels as follows:

1) Application layer: corresponds to the OSI application, presentation and session levels. The application layer refers to how the most common programs communicate with other programs over a network. Some specific programs run at this level. They provide services that work directly with user applications. These programs and their corresponding protocols include **HTTP** (hypertext transfer protocol), **FTP** (file transfer protocol), **SMTP**, **SSH**, **DNS**, among others.

OSI

OSI is the Open Systems Interconnection reference protocol, which was launched in 1984 and became the descriptive network model created by the ISO (International Organisation for Standardisation).

2) Transport layer: matches the transport level of the OSI model. Transport layer protocols solve issues such as reliability and security, making sure data reaches its destination in the correct sequence. Within the TCP/IP protocol suite, transport protocols also determine the target application for the data. **TCP** is a reliable, connection-based transport mechanism. It provides a reliable stream of bytes, ensuring that data arrives complete, undamaged and in sequence. **UDP** is a connectionless datagram protocol. It is an unreliable protocol, which does not verify that packets reach their destination, nor does it guarantee that they arrive in sequence. UDP is typically used for streaming applications where the timely delivery of packets is more important than reliability, or for simple request/response applications such as DNS, where the header overhead required for reliability is disproportionate to the size of the packets.

3) Inter-network layer: this refers to the OSI model's network layer. When originally defined, this layer dealt with the issue of transporting packets through a simple network.

4) Link layer: this is not actually part of the TCP/IP stack, but rather the method used to pass packets from one device's Internet layer to that of another device. This process can be controlled by both software and hardware.

3.2. IPv6 or IPng (next generation Internet Protocol)

This is the latest version of the Internet Protocol (IP), designed to replace the current version (known as **IPv4**). It was designed by Steve Deering and Craig Mudge, and adopted by the Institute Engineering Task Force (IETF) in 1994. The new version removes unused IP protocol functions and adds new ones. Here are some of the most important features of IPv6:

- **Increased addressing capacity**. One of IPv4's main shortcomings was its limited addressing capacity (2^{32}). The new addresses, consisting of 16 bytes, provide much higher addressing capacity, sufficient to prevent address allocation collapse: 2^{128}, approximately 3.4×10^{38}. In addition, and for the same reason, IPv4 cannot assign public addresses to all users or devices, without which end-to-end services cannot function (e.g., voice and video over IP).

- **Built-in security through IPSec**. This is a suite of secure network protocols that authenticates and encrypts data packets to provide secure encrypted communication between two computers over an Internet Protocol network. It is used in virtual private networks (known by the acronym VPN).

- **Mobility**. The possibility for a node to maintain its IP address, despite its mobility.

- **Self-configuration**. The new protocol also includes, as standard, a feature that allows the host to self-configure its interfaces and connect to the network.

Other key benefits include a new domain name system (DNS) that can be easily scaled up to new features, IPv6 tunnels in IPv4 (allowing machines with IPv6 installed to communicate with each other over an IPv4 network), and new types of addresses:

- **Unicast**. A packet sent to this type of address will only reach the interface identified by that address (equivalent to current IPv4 addresses).

- **Anycast**. The packet will reach *one* (the closest according to the addressing protocol) of the interfaces identified by the address of the group.

- **Multicast**. The packet will reach *all* addresses of the interfaces in the group (equivalent to IPv4 broadcast addresses).

4. Network configuration on computers (client/server)

The client/server concept encompasses other aspects that will not be discussed here. However, the computer acting as the server is the one that receives requests from other computers (the clients), which are usually connected to the same network.

To work in a client/server environment, clients must run network software on the workstation's operating system. Meanwhile, the server also runs its software, waiting to receive requests from workstations that want to access its services. This flow of information requires servers and clients to use the same communication protocol.

4.1. Workstation configuration

The steps to connect a workstation to the network are briefly described below. This operation is highly dependent on the system and protocol chosen, so all instructions provided should be considered general guidelines.

1) Network card driver installation and configuration

The first step is to install and configure the NIC drivers on the workstation.

Plug and Play card

If the network card is Plug and Play, it will be configured automatically.

In such cases, particularly when the cards come from different manufacturers, driver installation and configuration can be a complicated task, with I/O (input/output) and interruption conflict issues with other NICs or other system resources.

Note

One workstation may have more than one NIC!

2) Communication protocol selection and configuration

Some protocols, such as SPX/IPX for Novell or Apple Talk for Macintosh, are now obsolete due to new technologies. However, as mentioned above, the most common protocol is TCP/IP, without which it is impossible to access the Internet.

3) Client installation and configuration

This is where the applications that will use the network resources are installed, such as a file server that can serve FTP or file sharing clients through network drives, or a web or application server.

4) Other configurable aspects

At this point, other aspects can be configured, such as the following:

- Access control:
  - By resources: provides a password for each shared resource.
  - By users: specifies the users and groups that have access to each of the shared resources.

- File sharing:
  - Read permission.
  - Full.
  - Read or full permission depending on password.

- Printer sharing.

- Machine identification (this will be the name that appears on the network).

- Workgroup to which the machine belongs.

Printer drivers

The printer must be installed and have the necessary drivers on the connected computer. The workstations that will use it will also need to have the drivers installed.

Summary

Computer networks make better use of system resources. This module covered the components that form part of a network and some guidelines to help administrators choose these components and connect them to each other. Once the network is physically installed, the computers need to speak the same 'language'. In other words, they need to have the same communications protocol (the most commonly used is TCP/IP), which is installed as part of the workstation configuration. Despite the diversity of networks, protocols and network operating systems, these steps must always be taken in one way or another, although how they are done can vary greatly.

Finally, once the network is operational, the administrator must not forget to maintain the networks. They require maintenance to maintain the network, including creating and managing the user environment, monitoring the network, updating software, detecting attacks, etc.

Activities

1. Let's assume that you have access to a computer network. Answer the following questions:

a) Find and physically identify all the components that make up the network.
b) What topology has been used to design it?
c) What communication protocols are used?
d) How are the workstations configured?
e) What monitoring software is used?
f) Find and identify the security components (software and hardware).

2. If you do not have access to a computer network, make a list and describe all the components involved in an Internet connection via an Internet service provider:

Home ↔ Internet service provider ↔ Internet

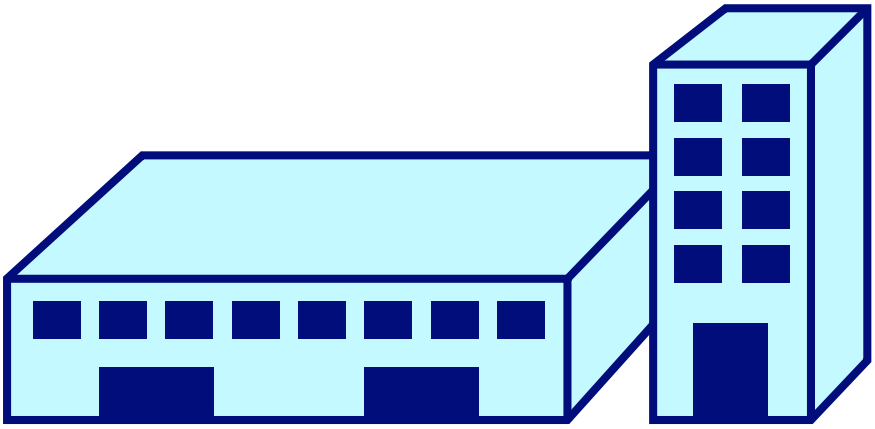
Another factor to bear in mind (which will also be discussed in the network security module) is that, for legal reasons (e.g. billing), an Internet provider should record the IP addresses it dynamically assigns to users, together with the phone number used to connect and the time interval between uses.

Self-evaluation

1. Fill in each of the boxes in the table below with one of the following options: *low/moderate/high*.

	Twisted pair	Coaxial	Fibre optic
Cost			
Bandwidth			
Length			
Interference			
Reliability			

2. Design and set up a network for a four-storey office building and a factory that uses a lot of motors. Plan the electrical cable layout to power the motors, then list the communication devices you'd need to install in both the factory and the office building to set up a local network connecting the offices to the workstations in the factory.



3. Identify which of the following features cannot be attributed to any star topology:

a) All workstation connect to a central element.
b) When a workstation sends a message, it always reaches all workstation on the network.

c) The topology is resistant to workstation crashes.
d) The central device can be active or passive.

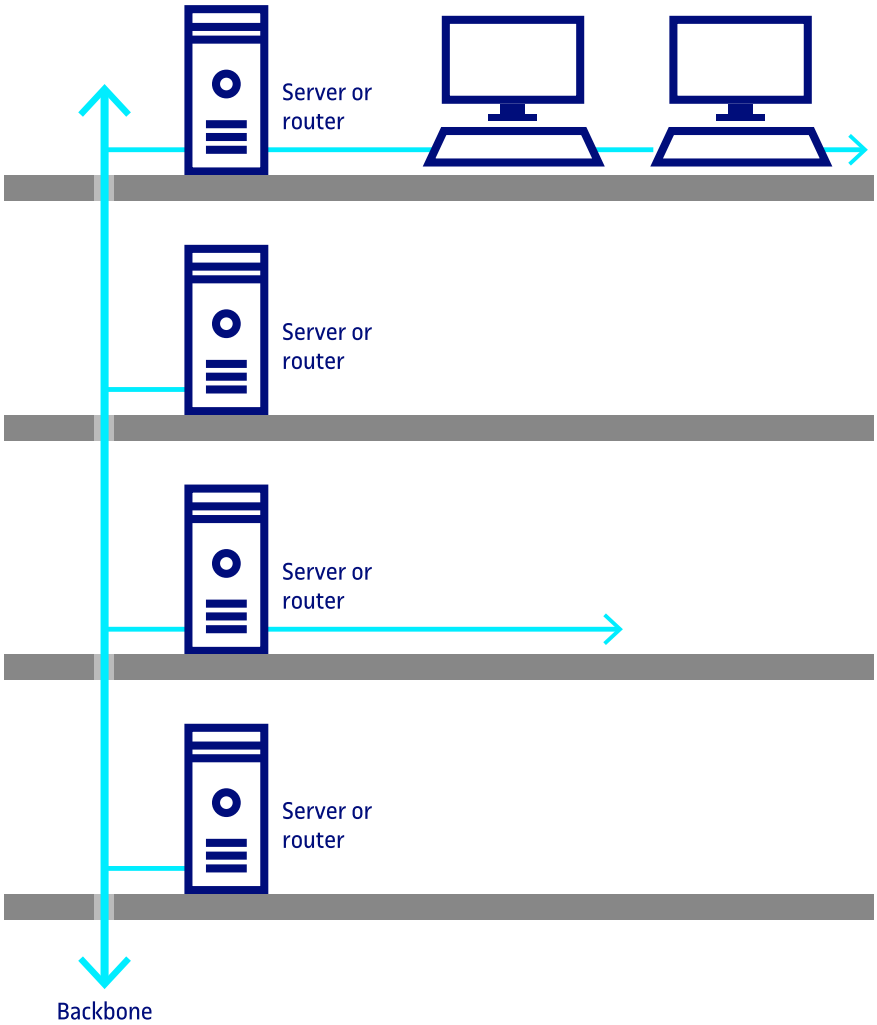
Answer key

Self-evaluation

1.

	Twisted pair	Coaxial	Fibre optic
Cost	Low	Moderate	High
Bandwidth	Moderate	High	Very high
Length	100 m	1 km	A few km
Interference	Low	Very low	None
Reliability	High	High	Very high

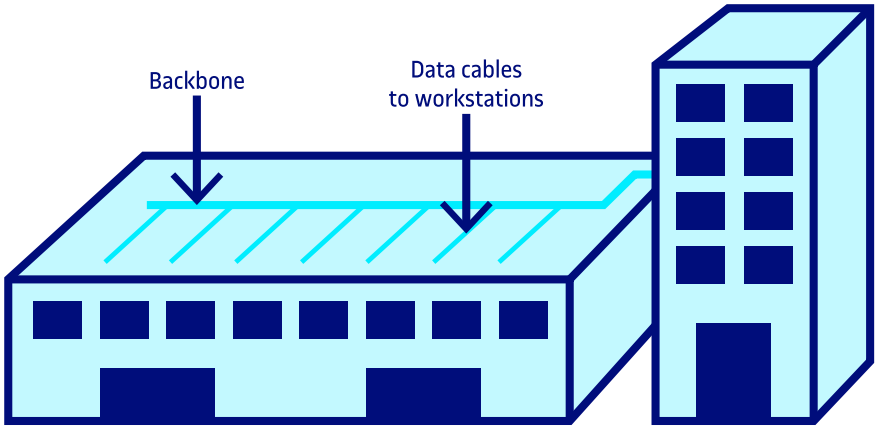
2.



The cabling in the facility should also follow a backbone layout, with branches at the necessary points. The basic design considerations are as follows:

- Cabling distances must not exceed the permitted limits. If they do, signal repeaters must be installed.
- Electrical interference must be taken into account, which means that electricity and data cannot travel through the same routes.

- If there are switch cabinets, vibration protection and power supply issues should be considered.



3. b

Glossary

10Base-2 Thin coaxial cable with a transmission speed of 10 Mbps. It can handle up to thirty workstations in segments of up to one hundred and eighty-five metres long.
syn.: thin wire

10Base-5 Thick coaxial cable with a transmission speed of 10 Mbps. It can handle up to one hundred workstations in segments ranging from a few metres to five hundred metres long.
syn.: thick wire

10Base-T UTP twisted pair cable with a maximum segment length of 100 metres in a star physical topology.

100Base-T UTP twisted pair cable with transmission speeds of 100 Mbps.
syn.: Fast Ethernet

backbone Set of main cables connecting the segments of a local network. These are usually high-speed connections (e.g. fibre optics).

DHCP See **dynamic host configuration protocol**.

dynamic host configuration protocol TCP/IP protocol that supports dynamic IP address assignment.

fast Ethernet See **100Base-T**.

firewall Any device (hardware or software) that prevents unauthorised users from accessing a specific machine.

hub A device for sharing a communication line between several computers. It distributes all the information it receives to all connected devices.

IEEE See **Institute of Electrical and Electronic Engineers**.

Institute of Electrical and Electronic Engineers An organisation founded in 1980 that developed the IEEE 802.x standards, defining standard operating procedures for local area networks.
acronym: IEEE

network interface card Interface card that connects the workstation to the network.

NIC See **network interface card**.

QoS See **Quality of service**.

router A device that manages packet traffic from outside the network to the inside (and vice versa). It can act as a firewall. It can filter and find the optimal route for packets.

switch A device that manages network traffic flow by considering each packet's destination address. Switches can determine which devices are connected to their ports and redirect information only to the destination port, unlike hubs, which send information indiscriminately.

thick wire See **10Base-2**.

thin wire See **10Base-5**.

wireless local area network Local wireless telecommunications network based on radio or infrared waves.

WLAN See **wireless local area network**.

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